

HSPI FPGA Verification Description

The FPGA verification process is as follows: FPGA sends data to CH569 in 16-bit mode, and CH569 prints characters to indicate the reception result according to the reception situation.

Sender: FPGA (model: EP4CE115F23I7)

Receiver: CH569

Frequency: 120MHz

Bit width: 16 bits

Data packet length: 512 bytes

HSPI transceiver mode: normal mode

Notes:

1. The distance between FPGA and CH569 should be as short as possible. Long wire connection will not work during debugging, no matter how constrained.
2. FPGA and CH569 share several GND signals.
3. Constrain the data alignment relationship in the Sdc file.
4. In this verification, HTCLK does not use the output clock of EXM_HSPI, but the output clock of PLL is selected. The frequency is consistent with the HSPI transmission clock, and the phase difference is 225 (deg).
5. If it is only the FPGA transmission process, the HRACT of FPGA is fixed to GND.

